



SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE

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Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Assignment No. 14

Subject:	DevOps Lab
TCODE:	TE7950
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Branch	CSE, Batch (2023-27), L4
Academic Year & Semester	25 – 26, Sem VII
Date of Performance	23 rd July, 2026
Title of Assignment:	CI/CD: Jenkins Pipeline building Docker Image
Theory:	A Jenkins Pipeline automates the Continuous Integration and Continuous Deployment (CI/CD) process by executing a series of predefined stages such as source code checkout, build, test, and deployment. In this experiment, a Docker image was built automatically using Jenkins after fetching the application source code from a GitHub repository. The docker build command created a versioned Docker image using the Jenkins build number as the image tag, ensuring unique image versions for every pipeline execution. This integration of Jenkins and Docker enables consistent, repeatable, and automated application builds, reducing manual effort and improving software delivery efficiency. Commands Used • docker build -t my-jenkins-docker . • docker run -d --name myjenkins --restart unless-stopped -u root -p 8080:8080 -p 50000:50000 -v jenkins_home:/var/jenkins_home -v /var/run/docker.sock:/var/run/docker.sock my-jenkins-docker • docker exec -it myjenkins docker --version • docker exec -it myjenkins rm -rf /var/jenkins_home/workspace/Declarative-Pipeline-Demo • git add . • git commit -m "Add Dockerfile for Jenkins pipeline" • git push origin main • docker images

Implementation:

Step 1: Access and Edit Your Pipeline Configuration

Accessed Jenkins Pipeline Configuration — Opened the existing Declarative Jenkins Pipeline and navigated to the Pipeline configuration section to modify the pipeline script.

Step 2: Add the Build Docker Image Stage

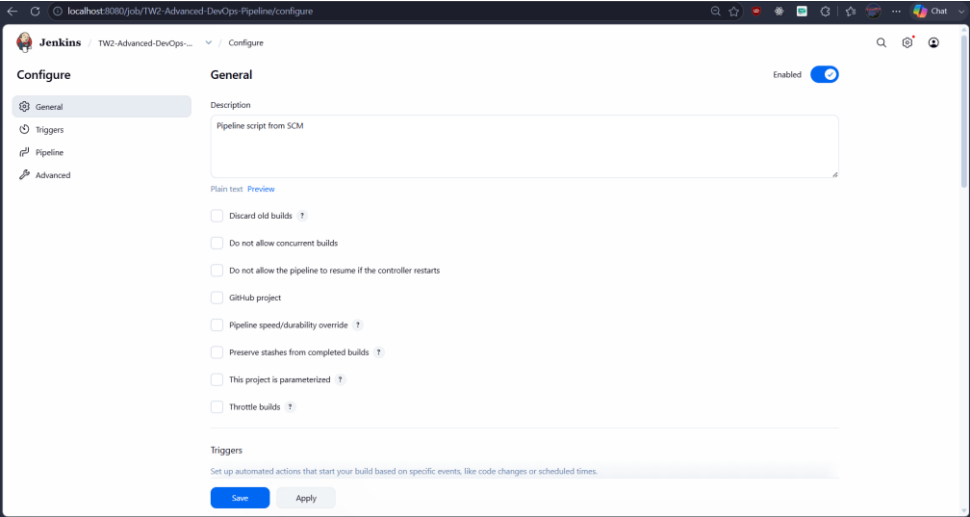
Added Docker Build Stage — Modified the existing Declarative Jenkins Pipeline by adding a new stage to build the Docker image using the docker build command with the Jenkins build number as the image tag.

Step 3: Run the Pipeline

Executed Jenkins Pipeline — Successfully executed the Jenkins Declarative Pipeline containing the Docker image build stage. The pipeline completed successfully, and the Docker image was built and tagged without errors.

Step 4: Verify the Docker Image

Verified Docker Image — Verified that the Docker image (my-flask-app:8) was successfully built by the Jenkins Pipeline and stored in the local Docker image repository using the docker images command.



```
PS C:\WINDOWS\System32> docker exec -u root -it jenkins bash
root@687c3418ae88:/# ls -l /var/run/docker.sock
srw-rw---- 1 root root 0 Aug 3 07:41 /var/run/docker.sock
root@687c3418ae88:/# chmod 666 /var/run/docker.sock
root@687c3418ae88:/# ls -l /var/run/docker.sock
srw-rw-rw- 1 root root 0 Aug 3 07:41 /var/run/docker.sock
root@687c3418ae88:/# exit
exit
PS C:\WINDOWS\System32> docker restart jenkins
jenkins
PS C:\WINDOWS\System32> docker exec -it jenkins bash
jenkins@687c3418ae88:/# docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED   STATUS    PORTS
687c3418ae88   jenkins/jenkins:lts   "/usr/bin/tini -- /u-"   16 minutes ago   Up 23 seconds   0.0.0.0:8080->8080/tcp, :::8080->8080/tcp, 0.0.0.0:50080->50080/tcp, :::50080->50080/tcp
6815a795734f   gcr.io/k8s-minikube/kicbase:v0.0.50   "/usr/local/bin/entr-"   46 minutes ago   Up 46 minutes   127.0.0.1:68765->22/tcp, 127.0.0.1:68763->2376/tcp, 127.0.0.1:68762->5000/tcp, 127.0.0.1:68764->8443/tcp, 127.0.0.1:68766->32443/tcp   minikube
jenkins@687c3418ae88:/#
```

```
Jenkins TW2-Advanced DevOps... #6

8175c193d46: Waiting
77bc405042c: Waiting
d77197981b22: Waiting
65956a0bc1c: Waiting
8175c193d46: Waiting
77bc405042c: Layer already exists
d77197981b22: Waiting
65956a0bc1c: Waiting
ef1cd8bdf467: Waiting
243f60c26961: Waiting
243f60c26961: Waiting
8175c193d46: Waiting
65956a0bc1c: Layer already exists
8175c193d46: Waiting
ef1cd8bdf467: Pushed
d77197981b22: Pushed
243f60c26961: Pushed
8175c193d46: Pushed
latest: digest: sha256:a829a9538148b0a0860f08bc0fde5401bfa7a74087342c8955cb518eac54fef size: 856
[Pipeline]
[Pipeline] // withCredentials
[Pipeline]
[Pipeline] // stage
[Pipeline]
[Pipeline] // withEnv
[Pipeline]
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS

Jenkins 2.568.1
```

```
PS C:\WINDOWS\system32> docker images

IMAGE                                ID                                DISK USAGE  CONTENT SIZE  EXTRA
-----                                -                                -            -            -
ad4r5h/flask-app:latest              a829a9538148                    1.62GB      417MB
alpine:latest                        28bd5f4e8b56d                   13MB        3.93MB
assignment-tw2-advanced-devops-flask-app:latest 71496c23d560c                   1.62GB      417MB
docker.n8n.io/n8nio/n8n:latest       cda6bafc7bb4                    2.47GB      387MB
gcr.io/k8s-minikube/kicbase:v0.0.50 b97874569ae9                    1.94GB      545MB
gcr.io/k8s-minikube/kicbase:sha256:eb4fec08e8ad70adf8e6436f195cc429825ffb85f95afcd5d8d9deb576f3e93 eb4fec08e8ad                    1.94GB      545MB
jenkins-agent-maven:latest           03fcc0aebfd9                    821MB       240MB
jenkins/inbound-agent:latest         6274f49f8642                    479MB       151MB
jenkins/jenkins:lts                  fu465e6cd140                    881MB       290MB
jenkins:2.60.3                       eeb4858eb65f                    1.08GB      310MB
my-flask-app:latest                   0d9d368293d9                    280MB       48.8MB
n8nio/n8n:latest                      cda6bafc7bb4                    2.47GB      387MB
postgres:15                          74e118c41804                    633MB       164MB
react-docker-app:latest              8b9560310983                    547MB       130MB

PS C:\WINDOWS\system32> docker rmi ad4r5h/flask-app:latest
Untagged: ad4r5h/flask-app:latest
Deleted: sha256:a829a9538148b0a0860f08bc0fde5401bfa7a74087342c8955cb518eac54fef
PS C:\WINDOWS\system32> docker pull ad4r5h/flask-app:latest
latest: Pulling from ad4r5h/flask-app
d77197981b22: Pull complete
243f60c26961: Pull complete
8175c193d46: Pull complete
ef1cd8bdf467: Download complete
Digest: sha256:a829a9538148b0a0860f08bc0fde5401bfa7a74087342c8955cb518eac54fef
Status: Downloaded newer image for ad4r5h/flask-app:latest
docker.io/ad4r5h/flask-app:latest
PS C:\WINDOWS\system32>
```

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